

PRACTICAL WEEK 3

TASK1: : Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

```
jhansi_priya@Jhansi-Priya: ~/  × + v - □ ×  
jhansi_priya@Jhansi-Priya:~$ mkdir practical3  
jhansi_priya@Jhansi-Priya:~$ cd practical3  
jhansi_priya@Jhansi-Priya:~/practical3$ nano process.c  
jhansi_priya@Jhansi-Priya:~/practical3$ gcc process.c -o process  
jhansi_priya@Jhansi-Priya:~/practical3$ ./process  
  
Parent Process  
PID = 408  
PPID = 291  
  
Child Process  
PID = 409  
PPID = 408  
jhansi_priya@Jhansi-Priya:~/practical3$ |
```

```
 jhansi_priya@Jhansi-Priya: ~/ × + v - □ ×  
GNU nano 7.2 process.c *  
#include <stdio.h>  
#include <unistd.h>  
#include <sys/types.h>  
#include <sys/wait.h>  
  
int main()  
{  
    pid_t pid;  
    pid = fork();  
    if (pid < 0)  
    {  
        printf("Process creation failed.\n");  
    }  
    else if (pid == 0)  
    {  
        printf("\nChild Process\n");  
        printf("PID = %d\n", getpid());  
        printf("PPID = %d\n", getppid());  
        sleep(5);  
    }  
    else  
    {  
        printf("\nParent Process\n");  
        printf("PID = %d\n", getpid());  
        printf("PPID = %d\n", getppid());  
        wait(NULL);  
    }  
    return 0;  
}
```

TASK2: Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

```

PPID = 475
jhansi_priya@Jhansi-Priya:~/practical3$ nano process.c
jhansi_priya@Jhansi-Priya:~/practical3$ gcc process.c -o process
jhansi_priya@Jhansi-Priya:~/practical3$ ./process

Parent Process
PID = 485
PPID = 291

Child Process
PID = 486
PPID = 485
jhansi_priya@Jhansi-Priya:~/practical3$ |

```

```

jhansi_priya@Jhansi-Priya: ~/p  X  jhansi_priya@Jhansi-Priya: ~  X  +  v
jhansi_priya@Jhansi-Priya:~$ ps -el
 F S   UID        PID    PPID    C  PRI   NI     ADDR   SZ    WCHAN    TTY          TIME CMD
 4 S    0           1         0  0   80    0      0     5432   -      -        ?           00:00:00 systemd
 5 S    0           2         1  0   80    0      0     795    -      -        hvc0        00:00:00 init-systemd(Ub
 0 S    0           7         2  0   80    0      0     799    -      -        hvc0        00:00:00 init
 4 S    0          53         1  0   79   -1     -1    10565   -      -        ?           00:00:00 systemd-journal
 4 S    0          99         1  0   80    0      0     6285   -      -        ?           00:00:00 systemd-udevd
 4 S   991        149         1  0   80    0      0     5336   -      -        ?           00:00:00 systemd-resolve
 4 S   996        159         1  0   80    0      0    22759   -      -        ?           00:00:00 systemd-timesyn
 4 S    0        165         1  0   80    0      0     1061   -      -        ?           00:00:00 cron
 4 S   101        166         1  0   80    0      0     2385   -      -        ?           00:00:00 dbus-daemon
 4 S    0        173         1  0   80    0      0     4496   -      -        ?           00:00:00 systemd-logind
 4 S    0        175         1  0   80    0      0    439026 -      -        ?           00:00:00 wsl-pro-service
 4 S   102        187         1  0   80    0      0    55629   -      -        ?           00:00:00 rsyslogd
 4 S    0        194         1  0   80    0      0      781   -      tty1       00:00:00agetty
 4 S    0        205         1  0   80    0      0    26758   -      -        ?           00:00:00unattended-upgr
 5 S    0        289         2  0   80    0      0      796   -      -        ?           00:00:00SessionLeader
 5 S    0        290        289  0   80    0      0      800   -      -        ?           00:00:00Relay(291)
 4 S  1000        291        290  0   80    0      0    1520 poll_s pts/0       00:00:00bash
 4 S    0        292         2  0   80    0      0    1675   -      -      pts/1       00:00:00login
 4 S  1000        335         1  0   80    0      0    5080   -      -        ?           00:00:00systemd
 5 S  1000        336        335  0   80    0      0    5291   -      -        ?           00:00:00(sd-pam)
 4 S  1000        360        292  0   80    0      0    1520 poll_s pts/1       00:00:00bash
 5 S    0        418         2  0   80    0      0      796   -      -        ?           00:00:00SessionLeader
 5 S    0        419        418  0   80    0      0      800   -      -        ?           00:00:00Relay(423)
 4 S  1000        423        419  0   80    0      0    1519 do_wai pts/2       00:00:00bash
 0 R  1000        436        423  0   80    0      0    2080   -      -      pts/2       00:00:00ns

```

```

481 jhansi_+ 20 0 9284 5752 3568 R 0.0 0.1
jhansi_priya@Jhansi-Priya:~$ top
top - 14:34:51 up 10 min, 1 user, load average: 0.00, 0.00, 0
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 z
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi,
MiB Mem : 7635.6 total, 7135.9 free, 500.6 used, 147.
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7135.

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM
1	root	20	0	21728	13248	9848	S	0.0	0.2
2	root	20	0	3180	2208	2072	S	0.0	0.0
7	root	20	0	3196	2124	2008	S	0.0	0.0
53	root	19	-1	50456	16220	15136	S	0.0	0.2
99	root	20	0	25140	6544	5204	S	0.0	0.1
149	systemd+	20	0	21344	13260	11132	S	0.0	0.2
159	systemd+	20	0	91036	8032	7060	S	0.0	0.1
165	root	20	0	4244	2748	2496	S	0.0	0.0
166	message+	20	0	9540	5476	4776	S	0.0	0.1
173	root	20	0	17984	8916	7868	S	0.0	0.1
175	root	20	0	1756104	13252	11016	S	0.0	0.2
187	syslog	20	0	222516	6000	4612	S	0.0	0.1
194	root	20	0	3124	2040	1900	S	0.0	0.0
205	root	20	0	107032	23344	13892	S	0.0	0.3
289	root	20	0	3184	1112	980	S	0.0	0.0
290	root	20	0	3200	1252	1108	S	0.0	0.0
291	jhansi_+	20	0	6080	5388	3668	S	0.0	0.1
292	root	20	0	6700	4740	3956	S	0.0	0.1
335	jhansi_+	20	0	20320	11644	9516	S	0.0	0.1
336	jhansi_+	20	0	21164	3592	1868	S	0.0	0.0
360	jhansi_+	20	0	6080	5340	3672	S	0.0	0.1
418	root	20	0	3184	1120	980	S	0.0	0.0
419	root	20	0	3200	1260	1108	S	0.0	0.0
423	jhansi_+	20	0	6076	5372	3648	S	0.0	0.1
489	jhansi_+	20	0	9284	5724	3560	R	0.0	0.1

```

jhansi_priya@Jhansi-Priya:~$ ps
PID TTY TIME CMD
423 pts/2 00:00:00 bash
498 pts/2 00:00:00 ps

```